

FINAL PROJECT OPTIMIZATION

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Introduction

This document presents the final project for the Optimization course. The main objective of the project is to determine an optimal weekly schedule for a community room of a neighborhood association, where different activities take place. We aim to design a schedule that specifies which activities should be held at each time slot, considering the limited resources of the community and trying to serve as many community members as possible.

To address this problem, we developed an optimization model, which is described in detail in the following sections. This model assigns each activity to a specific day, time, and duration within the week. The interesting part of this model is that it considers two complementary optimization objectives: maximizing the minimum attendance across all scheduled sessions and maximizing the total number of participants across all activities.

In the following parts, we will present the specific details of the problem, our formulation of the model, and discuss the solution obtained.

Problem to Solve

A community center offers several activities throughout the week for its members, including Yoga classes, Workshops, Meetings and Children Activities. These activities must be scheduled within a single room that is shared by all sessions. The room can only hold one activity at a time. The planning on the weekly schedule must consider that the community has limited resources and constraints. Every type of activity must be scheduled at least once a week, and not more than five times a week.

To do the formulation of the problem, every weekday is divided into two time periods: morning and afternoon, and every time period has 5 blocks that have 1 hour duration.

Every activity may require the presence of a supervisor. In this case, we establish that every activity needs one, but the problem is formulated taking this parameter into account. The community has two supervisors at their disposal, but they have specific availability through the week. Therefore, one supervisor cannot be assigned to more than one activity at the same time.

Also, the room has a maximum capacity that limits the number of participants that can attend an activity simultaneously. This capacity is 25 people. For every activity and time slot, there is an estimated demand that represents the number of people that wish to attend. The number of participants assigned to every session cannot exceed either the room capacity or the estimated demand.

Activity	Duration (hours)	Estimated Demand (people)
Yoga	2	15
Kids Class	2	20
Community Meeting	1	10
Crafts Workshop	1	10

The objective is to determine an optimal weekly schedule by deciding which activities are scheduled on which days and at what times, which supervisors are assigned to each activity, and how many participants attend each session. The schedule must satisfy all constraints, including room availability, non-overlapping activities, supervisor requirements, capacity limits, and minimum and maximum numbers of weekly sessions per activity.

This problem is addressed using two optimization criteria:

1. Maximize the minimum attendance along all scheduled sessions
2. Maximize the total number of participants served throughout the week.

Therefore, we try to reach a solution that both seeks to have a fair distribution and efficient schedule for the community center.

Problem Formulation

Sets

- Days (D): Represents the time horizon for a full week.

$$D = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$$

- Periods (P): Each day has two main time periods.

$$P = \{\text{Morning, Afternoon}\}$$

- Activities (A): List of activities.

$$A = \{\text{Yoga, Community Meeting, Kids Class, Crafts Workshop}\}$$

We have picked a set of generic activities for this model.

- Supervisors (S): The people in charge of supervising the activities.

$$S = \{\text{Supervisor 1, Supervisor 2}\}$$

- Hours in each time period (H_p): This set defines the specific hourly blocks that compose each period $p \in P$.

$$H_{\text{Morning}} = \{1, 2, 3, 4, 5\}$$

$$H_{\text{Afternoon}} = \{1, 2, 3, 4, 5\}$$

For this model, each period contains 5 blocks of 1h of duration.

- Tridimensional decision space: This set represents the complete structure where the model can place an activity.

$$\Omega = \{(d, p, h) : d \in D, p \in P, h \in H_p\}$$

Each element (d, p, h) represents a specific unit of time composed of an hourly block, a day of the week and a period of the day.

Parameters

We have defined the following parameters along with some temporary values in order to test the functioning of the model.

1. Room Availability ($RA_{d,p,h}$): A binary parameter that indicates if the venue is available for use at a specific time.

Set to 1 for all days, periods, and hours, assuming the venue is always available for booking.

2. Minimum Weekly Sessions ($S_{min,a}$): The minimum number of times an activity must be held during the week.

At least 1 session per week for every activity.

3. Maximum Weekly Sessions ($S_{max,a}$): The maximum number of times an activity can be held over the week.

A maximum of 5 sessions per week for every activity.

4. Activity Duration (Dur_a) The number of consecutive time slots (hours) that a single session of activity a occupies.

We have established the following values: Yoga: 2 slots, KidsClass: 2 slots, CommunityMeeting: 1 slot, CraftsWorkshop: 1 slot.

5. Supervision Requirement (NS_a): A binary parameter that indicates if activity a requires the presence of a supervisor.

All activities are set to 1 (require supervision).

6. Supervisor Availability ($SA_{s,d,p,h}$): A binary parameter representing the working schedule of each supervisor.

Set to 1 (available) for all times, except Sunday Afternoons, where it is set to 0 to represent staff rest time.

7. Venue Capacity (K) A scalar value representing the maximum number of people allowed in the room at the same time.

Fixed at 25 people.

8. Estimated Demand ($D_{a,d,p,h}$): The expected number of neighbors interested in attending activity a at a specific time.

Yoga: 15 people, KidsClass: 20 people, Others: 10 people

Variables

1. $X_{a,d,p,h}$: Specific time slot (d, p, h) where activity a begins. If an activity has a duration of 2 hours, this variable is only 1 at the starting hour.

Type: Binary

2. $n_{a,d,p,h}$: Represents the number of neighbors attending a specific session of an activity.

Type: Integer

3. $y_{a,d,p,h,s}$: Assigns supervisors to specific activity sessions. It ensures that every activity has the required staff as defined by the NS_a parameter

Type: Binary

4. $R_{a,d,p,h}$: Auxiliary variable that tracks if the room is occupied. This is necessary because, while X only marks the start, R remains 1 for the entire duration of the activity. This is relevant in preventing two activities from overlapping.

Type: Binary

5. $Z_{\min}$: minimum number of persons among all sessions. It is an auxiliary variable used in the objective function to ensure that the least attended session is still as full as possible.

Type: Integer

Constraints

1. Valid start time: Calculates whether an activity a starting at hour h has enough time to finish before the period p (Morning or Afternoon) ends. It acts as a "filter" that other constraints use to know which time slots are worth considering.

$$h + \text{Dur}_a - 1 \leq \max(H_p)$$

2. Invalid start: Takes the result of the valid start time rule and makes sure that if the activity doesn't fit, the decision variable X must be zero.

$$X_{a,d,p,h} = 0 \quad \forall (a, d, p, h) \notin \text{ValidStart}$$

3. Activity Running link: defines the active state of the venue by linking the starting variable (X) with the occupancy variable (R). It ensures that if an activity starts at time t , all the following slots within its duration are marked as occupied.

$$R_{a,d,p,h} = \sum_{t \in H_p: t \leq h \leq t + \text{Dur}_a - 1} X_{a,d,p,t}$$

4. No overlapping activities in the room: This constraint ensures that the sum of all "Running" activities (R) never exceeds the available physical capacity (one room).

$$\sum_{a \in A} R_{a,d,p,h} \leq 1$$

5. Room availability: While the previous constraint handles physical capacity, this constraint handles operational policy. It links the activity schedule to the center's official opening hours. By using the parameter $RA_{d,p,h}$ the administration can CLOSE the room for specific blocks (e.g., for cleaning, maintenance, or public holidays) by setting the parameter to 0. When the room is closed, this constraint overrides the physical capacity and forces all activity variables to zero.

$$\sum_{a \in A} R_{a,d,p,h} \leq RA_{d,p,h} \quad \forall (d, p, h) \in \Omega$$

6. Mandatory Supervision Assignment: This constraint ensures that every scheduled activity is properly staffed according to its specific safety requirements.

$$\sum_{s \in S} y_{a,d,p,h,s} \geq NS_a \cdot X_{a,d,p,h} \quad \forall a, d, p, h$$

7. Supervisor Availability for Full Duration: Make sure that a supervisor assigned to an activity is available to oversee it from start to finish.

$$y_{a,d,p,h,s} \leq SA_{s,d,p,k} \quad \forall a, (d, p, h) \in \Omega, s \in S, \quad \forall k \in \{h, \dots, h + \text{Dur}_a - 1\}$$

If the supervisor is unavailable at any point during the activity's duration ($SA = 0$), the variable y is forced to be 0. This ensures that the model only selects supervisors who can commit to the entire length of the session.

8. Supervisor Non-Overlapping: Ensures that no supervisor is double-booked across multiple activities at the same time.

$$\sum_{a \in A} \sum_{t \in H_p: t \leq h \leq t + \text{Dur}_a - 1} y_{a,d,p,t,s} \leq 1 \quad \forall s, d, p, h$$

9. Capacity of the room: Number of attendees cannot exceed capacity, and must be zero if the activity is not scheduled

$$n_{a,d,p,h} \leq K \cdot X_{a,d,p,h} \quad \forall a, d, p, h$$

10. Demand limitation: You cannot have more attendees than the estimated demand for a specific activity and time slot.

$$n_{a,d,h_p} \leq \text{demand}_{a,d,h_p} \quad \forall a \in A, \forall d \in D, \forall h_p \in H$$

11. Minimum and maximum number of sessions per activity.

- Minimum sessions

$$\sum_{(d,p,h) \in \Omega_{\text{valid}}} X_{a,d,p,h} \geq S_{\min,a} \quad \forall a \in A$$

- Maximum sessions

$$\sum_{(d,p,h) \in \Omega_{\text{valid}}} X_{a,d,p,h} \leq S_{\max,a} \quad \forall a \in A$$

12. Fairness (Min-Max): This constraint defines the auxiliary variable Z , which represents the minimum attendance level among all scheduled activities. We use Big-M formulation used to achieve a balanced and fair distribution of participants.

$$Z \leq n_{a,d,p,h} + M \cdot (1 - X_{a,d,p,h}) \quad \forall a, d, p, h$$

Where M is a large constant , set to the maximum capacity of the room ($K = 25$).

The objective of this constraint is to obtain the minimum number of persons amongst all sessions. In the objective function, by maximizing Z , we are effectively trying to maximize the attendance of the least popular.

13. Non-Negativity

$$X_{a,d,p,h}, R_{a,d,p,h}, Y_{a,d,p,h,s} \in \{0,1\}$$

$$n_{a,d,p,h} \in \mathbb{Z}^+ \cup \{0\}$$

$$Z \geq 0$$

Objective Function

To achieve a balance between high attendance and still include activities for a wide range of audience, the model utilizes a sequential optimization approach.

1. Maximizing the Minimum (Fairness):

- Function: Max Z
- First we want to maximize the attendance of the least popular session. By maximizing the variable Z , we are forcing the model to create a schedule where every activity, even the less popular ones, reaches a respectable level of participation.

2. Maximizing Total Volume (Efficiency)

- Function: Max $\sum_{a,d,p,h} n_{a,d,p,h}$
- After finding the optimal fairness value Z^* , the first objective is deactivated, and the second is activated. The model will now maximize the total number of neighbors that come to all sessions, always within the constraints established by the first step.

Analysis and interpretation of the results

Using the results obtained we can use them to create the full calendar for the week. In it we can see how each class was allocated to each time block.

DAY	PERIOD	HOUR	ACTIVITY	ATTENDEES	SUPERVISOR
Monday	Afternoon	1	Yoga	15	Supervisor1
Monday	Afternoon	3	CommunityMeeting	10	Supervisor1, Supervisor2
Tuesday	Morning	2	Yoga	15	Supervisor2
Tuesday	Afternoon	1	CraftsWorkshop	10	Supervisor1
Tuesday	Afternoon	2	Yoga	15	Supervisor1
Tuesday	Afternoon	4	Yoga	15	Supervisor1
Thursday	Afternoon	1	CraftsWorkshop	10	Supervisor1, Supervisor2
Thursday	Afternoon	2	KidsClass	20	Supervisor2
Friday	Morning	2	KidsClass	20	Supervisor1
Friday	Morning	5	CommunityMeeting	10	Supervisor2
Friday	Afternoon	1	CraftsWorkshop	10	Supervisor1
Friday	Afternoon	3	Yoga	15	Supervisor2
Saturday	Morning	1	CraftsWorkshop	10	Supervisor2
Saturday	Morning	2	CommunityMeeting	10	Supervisor1
Saturday	Morning	5	CommunityMeeting	10	Supervisor1
Saturday	Afternoon	2	KidsClass	20	Supervisor2
Saturday	Afternoon	4	KidsClass	20	Supervisor1, Supervisor2
Sunday	Morning	1	CommunityMeeting	10	Supervisor2
Sunday	Morning	2	KidsClass	20	Supervisor2
Sunday	Morning	5	CraftsWorkshop	10	Supervisor1

It seems that some activities don't meet their required time slots, such as yoga which must be two hours long. This due to the code used to print the table, therefore it should only be used to get a general idea of how the activities are being distributed instead of a concrete calendar.

We can see that there are a lot of time intervals with no allotted classes, for example Wednesday is missing altogether. This can probably be attributed to the fact that there were enough slots to allow all classes to meet their maximum number of sessions, which is 5. Therefore, there are a lot of slots that have not been filled.

In conclusion, we have developed an optimization model with 2 different objective functions to compare the different results with different criteria. It is important to mention that all variables and parameters were defined using fictitious values and, in the future, we plan to use real data when continuing this work with APS.

The two different objective functions explored corresponded to different planning perspectives. In the first one, the objective was to maximize the minimum number of participants across all scheduled sessions. This objective prioritizes fairness and balance, ensuring that no session is run with very low attendance. In the second one, the objective was to maximize the total number of participants without imposing any requirement on how evenly they are distributed across activities. This second objective function lead to the same total attendance but allowed some sessions to be much smaller with the fictitious values we used, this suggests that since the total number of participants was the same in both objective functions balancing the number of participants across all activities doesn't (with this data) result in any loses in the total number of participants.

This phenomenon probably occurs as a consequence of our specific fictitious data we used for the model. We can see that room capacity and supervisor availability did not yet reach their

saturation points. Consequently, these resources did not act as binding constraints, allowing the model to satisfy the fairness objective without compromising the global maximum.

```
First objective function  
Z = optimum minimum = 10.0  
Total people assisting = 275.0
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Second objective function  
Z = -0.0  
Total people asisting = 275.0
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